

$$\equiv T = \{\{a, 1-b\}, \{1-a, b\}\}$$

In[3]:=  $T = \{\{a, 1 - b\}, \{1 - a, b\}\}$

$$\text{Out[3]} = \begin{pmatrix} a & 1-b \\ 1-a & b \end{pmatrix}$$

$$\equiv \text{Eigenvalues}[T]$$

In[4]:=  $\text{Eigenvalues}[T]$

$$\text{Out[4]} = \{1, a + b - 1\}$$

## ▣ Eigenvectors[T]

Eigenvectors[T]

$$6] = \begin{pmatrix} -\frac{1-b}{a-1} & 1 \\ -1 & 1 \end{pmatrix}$$

Prove the following statement: If  $T$  is a  $2 \times 2$  regular transition matrix then  $T$  has a steady state vector  $\vec{x}_*$  and furthermore for any initial vector  $\vec{x}_0$  we have  $\lim_{n \rightarrow \infty} T^n \vec{x}_0 = \vec{x}_*$ .

A  $2 \times 2$  Stochastic matrix is of the form:

$$T = \begin{bmatrix} a & 1-b \\ 1-a & b \end{bmatrix}, \text{ where } 0 \leq a \leq 1 \text{ and } 0 \leq b \leq 1.$$

Note that if  $a=b=0$ , then  $T = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$  and  $T^n = \begin{cases} T & n \text{ odd} \\ I & n \text{ even} \end{cases}$ . Hence

$T$  is not regular. Also, if  $a=b=1$ , then  $T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ , which is clearly not regular. Therefore, we can ignore these two cases.

To compute  $T^n$ , we first diagonalize  $T$ . We can compute the eigenvalues and the eigenvectors to get:

Eigenvalues :  $\lambda_1 = 1$  ,  $\lambda_2 = a+b-1$

Eigenvectors :  $\vec{x}_1 = \begin{bmatrix} b-1 \\ a-1 \end{bmatrix}$  ,  $\vec{x}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

Then, we diagonalize:

$$T = P D P^{-1} = \begin{bmatrix} b-1 & -1 \\ a-1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & a+b-1 \end{bmatrix} \begin{bmatrix} b-1 & -1 \\ a-1 & 1 \end{bmatrix}^{-1}$$

$$T^n = P D^n P^{-1} = \begin{bmatrix} b-1 & -1 \\ a-1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & (a+b-1)^n \end{bmatrix} \frac{1}{a+b-2} \begin{bmatrix} 1 & 1 \\ 1-a & b-1 \end{bmatrix}$$

Now,  $\lim_{n \rightarrow \infty} (a+b-1)^n = 0$ , since  $0 < a+b < 2$  and  $-1 < a+b-1 < 1$ .

$$\begin{aligned} \text{Therefore: } \lim_{n \rightarrow \infty} T^n &= \frac{1}{a+b-2} \begin{bmatrix} b-1 & -1 \\ a-1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1-a & b-1 \end{bmatrix} \\ &= \frac{1}{a+b-2} \begin{bmatrix} b-1 & -1 \\ a-1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \\ &= \frac{1}{a+b-2} \begin{bmatrix} b-1 & b-1 \\ a-1 & a-1 \end{bmatrix} \end{aligned}$$

Now, if the initial probability vector is  $\vec{x}_0 = \begin{bmatrix} p \\ 1-p \end{bmatrix}$ , then

$$\lim_{n \rightarrow \infty} T^n \vec{x}_0 = \frac{1}{a+b-2} \begin{bmatrix} b-1 & b-1 \\ a-1 & a-1 \end{bmatrix} \begin{bmatrix} p \\ 1-p \end{bmatrix}$$

$$= \frac{1}{a+b-2} \begin{bmatrix} b-1 \\ a-1 \end{bmatrix} \rightarrow$$

$$= \frac{1}{a+b-2} \vec{x}_1$$

Note that this is the probability vector corresponding to eigenvector  $\vec{x}_1$  with eigenvalue one.

$$\vec{x}_1 = \frac{1}{a+b-2} \begin{bmatrix} b-1 \\ a-1 \end{bmatrix}$$

## Summary

- (I)  $T_{2 \times 2}$  is a regular stochastic matrix.
- (II)  $\vec{x}_0$  is an initial probability vector.
- (III)  $\vec{x}_1$  is a probability eigenvector of  $T$  corresponding to eigenvalue one.
- (IV) The steady state vector is  $\vec{x}_1$ :  $\lim_{n \rightarrow \infty} T^n \vec{x}_0 = \vec{x}_1$ . This means to find the steady state vector of  $T_{2 \times 2}$ , we can find the eigenvector corresponding to eigenvalue one and normalize it.

Question: How do we know that  $T$  is diagonalizable?

Note that  $\vec{x}_1 = \begin{bmatrix} b-1 \\ a-1 \end{bmatrix}$  and  $\vec{x}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$  are two linearly independent eigenvectors.

Since  $-1 \leq b-1 \leq 0$  and  $-1 \leq a-1 \leq 0$ , hence  $\vec{x}_1 \neq C\vec{x}_2$ , where  $C \in \mathbb{R}$ .

Question: What if  $T_{2 \times 2}$  is not invertible, would the result above still hold?

$$\det T = a+b-1=0 \rightarrow b=1-a \rightarrow T = \begin{bmatrix} a & a \\ 1-a & 1-a \end{bmatrix}$$

Eigenvalues:  $\lambda_1 = 1$  and  $\lambda_2 = 0$

Eigenvectors:  $\vec{x}_1 = \begin{bmatrix} a \\ 1-a \end{bmatrix}$  and  $\vec{x}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Note that you can simply replace  $b$  by  $1-a$  and all the rest of the steps follow.

---

## Problem

Alice always eats lunch either at restaurant A or restaurant B. Suppose that

- Initially she is equally likely to eat at each restaurant.  $\rightarrow \vec{S}_0 = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$
- She never eats at A two days in a row.
- If she eats at restaurant B one day, then the next day she is three times as likely to eat at B as at A.

Calculate the probabilities of her eating at A and B for the first week. What do you notice?

(transition matrix)  $T = \begin{matrix} & \underbrace{\text{today}} \\ & \begin{matrix} A & B \end{matrix} \\ \begin{matrix} A \\ B \end{matrix} & \begin{bmatrix} 0 & \frac{1}{4} \\ 1 & \frac{3}{4} \end{bmatrix} \end{matrix} \left\{ \begin{matrix} A \\ B \end{matrix} \right\} \text{next day}$

$$\vec{S}_1 = T\vec{S}_0 = \begin{bmatrix} 0.125 \\ 0.875 \end{bmatrix}$$

$$\vec{S}_2 = T\vec{S}_1 = \begin{bmatrix} 0.2188 \\ 0.7812 \end{bmatrix}$$

$\vdots$

$$\vec{S}_6 = T\vec{S}_5 = \begin{bmatrix} 0.2001 \\ 0.7999 \end{bmatrix}$$

$$\vec{S}_7 = T\vec{S}_6 = \begin{bmatrix} 0.2001 \\ 0.7999 \end{bmatrix}$$

The state vectors seem to be converging to  $\begin{bmatrix} 0.2 \\ 0.8 \end{bmatrix}$ .

Also, we tried changing the initial state vector and it seemed that the transition matrix above seems to cause the state vectors to converge to  $\begin{bmatrix} 0.2 \\ 0.8 \end{bmatrix}$  independent of the choice of  $\vec{S}_0$ .

## Existence of a unique steady state

If the transition matrix  $T$  of a Markov Chain is a regular stochastic matrix, then it has a unique steady state.

### Theorem

If an  $n \times n$  regular matrix  $T$  is the transition matrix of a Markov chain, then the steady state vector can be found by solving  $S = TS$ .

### Problem

Does the matrix  $T = \begin{bmatrix} 0 & \frac{1}{4} \\ 1 & \frac{3}{4} \end{bmatrix}$  have a unique steady state vector? If so, find the steady state vector.

- (I)  $T$  is a stochastic matrix, since the sum of entries in each column is equal to one  
(II)  $T$  is a regular matrix, since  $T^2 = \begin{bmatrix} 0 & \frac{1}{4} \\ 1 & \frac{3}{4} \end{bmatrix} \begin{bmatrix} 0 & \frac{1}{4} \\ 1 & \frac{3}{4} \end{bmatrix} = \begin{bmatrix} \frac{1}{4} & \frac{3}{16} \\ \frac{3}{4} & \frac{13}{16} \end{bmatrix}$  and all the entries of  $T^2$  are positive.  
(I), (II)  $\longrightarrow T$  has a unique steady state vector.

$$T\vec{S} = \vec{S} \text{ or } (T - I)\vec{S} = \vec{0} \xrightarrow{\text{Solve for } \vec{S} = \begin{bmatrix} a \\ b \end{bmatrix}} \left[ \begin{array}{cc|c} -1 & \frac{1}{4} & 0 \\ 1 & -\frac{1}{4} & 0 \end{array} \right] \left[ \begin{array}{cc|c} 1 & -\frac{1}{4} & 0 \\ 0 & 0 & 0 \end{array} \right] \vec{S} = \begin{bmatrix} \frac{t}{4} \\ t \end{bmatrix} \quad t \in \mathbb{R}$$

Now, remember that entries of the state  $\vec{S}$  represent Alice's probabilities of eating at restaurant A and B, hence they must add up to one: (See the problem.)

$$\frac{t}{4} + t = 1 \longrightarrow \frac{5t}{4} = 1 \longrightarrow t = \frac{4}{5} \longrightarrow \text{Steady state: } \vec{S} = \begin{bmatrix} \frac{1}{5} \\ \frac{4}{5} \end{bmatrix} = \begin{bmatrix} 0.2 \\ 0.8 \end{bmatrix}$$

Note that this is the same result that we got numerically earlier on.